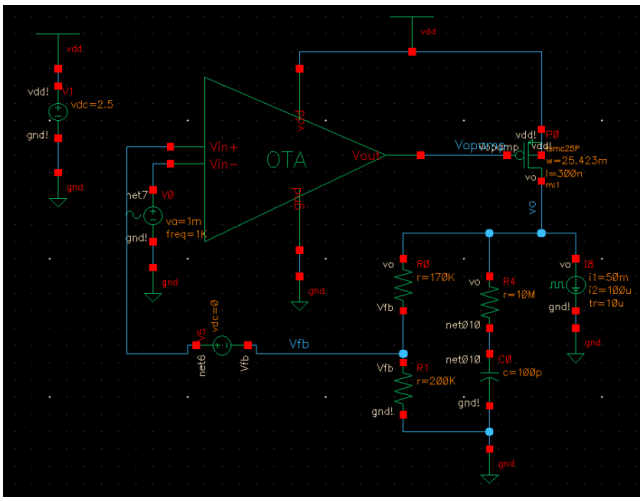
ANALOG IC DESIGN

2.3 V CMOS low-dropout regulator

Ruthvik Reddy Anthireddigari / Arizona State University / Fall 2025

An OTA drives a PMOS pass device through a feedback loop. With a 2.5 V supply, the simulated output varied by 8 mV across four DC load conditions.

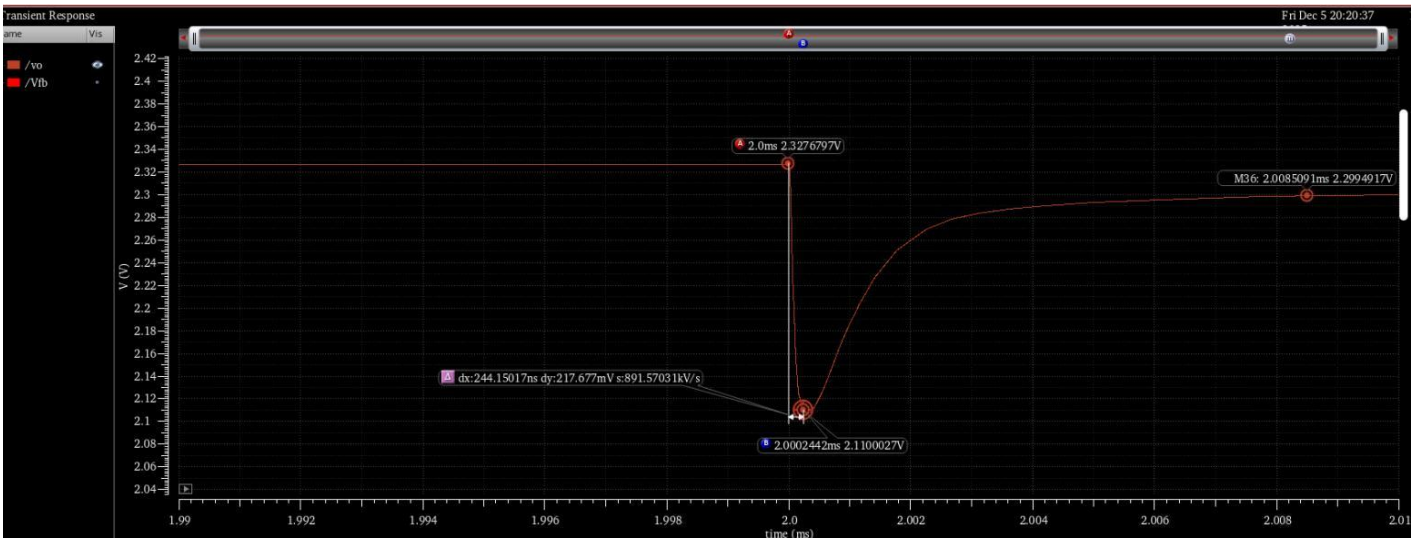
Closed-loop schematic



DC load sweep

Load current	Output voltage
0.1 mA	2.320 V
1 mA	2.323 V
10 mA	2.319 V
50 mA	2.315 V

Response to a 0.1 to 50 mA load step



The load increase briefly pulls the output down. Feedback adjusts the pass-device drive to restore regulation.

EEE 433/591 final project, pp. 8, 11-13, 16. DC values and waveforms are simulation results.